Take-Home Exam, Advanced Microeconomics

Problem 1: Overworked Americans – Q1, Q2, Q4, Q5, Q7

Alessandro Caggia

Bocconi

May 2026

Plan and storyline. Q1 – there is an inconsistency in the data. The US–EU hours-per-worker gap is real (panels (a)/(b)) but its cross-country relation to the tax wedge is loose: the 2023 OLS slope is signed correctly (−9.9 hrs/pp) but $R^2 = 0.16$ and the residual SD is ≈ 158 hours – one third of the entire DE–US gap (panel (c)). Taxes account for some of the cross-country variation, not most of it. **Q2 – the calibrated model fits, but only at an α above what micro evidence delivers; at the micro-consistent α, taxes explain only half the gap.** The Prescott log–log model gives $h^*(\tau) = (1-\tau)/(\alpha + 1 - \tau)$. At $\alpha^{\text{macro}} = 1.32$ (calibrated to $h^{US} = 0.33$) the model fits the cross-section. Restricting α to the micro-consistent $\alpha^{\text{micro}} = 0.28$ (such that $\varepsilon^M = 0.30$ at the US point), the Prescott-only prediction is a 13.4% US–EU log-drop vs. data 29.0% – so **taxes alone explain $\approx$46% of the cross-country hours gap**, leaving a residual 54% ($\sim$15.6 pp in logs) that the Prescott mechanism cannot account for; §4–§5 plus Q7 culture propose institutions and culture as candidate explanations and test whether they can absorb it. **Q4 – hours regulation:** under a low-η_0 RTM no-regulation baseline (the same machinery as §5 at small $\eta_0 \geq 0$, with $\eta_0 = 0$ nesting firm-power), a binding cap is welfare-improving iff $\bar{h} \in (2h^{\text{soc}} - h(\eta_0),\, h(\eta_0))$. The band is non-empty only for $\eta_0 < \eta^{\text{soc}} \approx 0.41$: *the cap and the union are substitutes.* Manning (2003) monopsony and Cahuc–Carcillo (2014) sticky contracts both map to low effective η_0. **Q5 – single union: canonical right-to-manage (Nickell 1982; LNJ 1991).** Union and firm Nash-bargain over wage w; firm sets $h = h^d(w)$ on labour demand. As η rises, w rises and the firm cuts h along its labour-demand curve. **The welfare verdict depends on the welfare criterion (the weight given to worker vs. firm surplus).** Under realistic Saez–Stantcheva (2016) inequality aversion ($\lambda = 0.30$; HSV 2017 + log utility), $\eta^* \approx 0.58$ – *a moderately strong union is welfare-optimal* (§5.5b). The equal-weight knife-edge $\lambda = 0$ gives the corner $\eta^* = 0$, but this assigns no value to the firm-to-worker redistribution and is not a defensible default. Applied to a two-sector economy with heterogeneous productivity, the *same* RTM machinery delivers the canonical single-sector story in the good sector (positive shock → union extracts wage rents → hours rise less than under free market) and an extra asymmetry in the bad sector (negative shock → wages cannot fall under Bewley 1999 / Holden 1994 downward rigidity → hours and profit absorb the entire shock). The good-sector outcome is the single-sector RTM result; the bad-sector outcome is what the same model produces once you allow the wage to be downward-sticky. **Good shocks pass through diluted; bad shocks pass through amplified. Q7 – welfare comparison: culture as expansion + layered counterfactuals.** The cultural primitive enters Q7 as a country-specific leisure-bonus expansion: worker payoff $V_w(h;\theta^c) = (1-\tau)wh - (\alpha/2)h^2 + \theta^c(1-h)$ with $\theta^{US} = 0$, $\theta^{EU} = 0.156$ (SW-anchored). Higher θ^c shifts h^* down AND adds direct welfare from leisure ($h^*_{US} = 0.30$, $h^*_{EU} = 0.222$, but $V_w^{EU} > V_w^{US}$ at first-best); refs Bowles–Park 2005, Olovsson 2009, Boppart–Krusell 2020, Stevenson–Wolfers 2008. Layered with the RTM bargain ($\eta^{EU} = 0.5$, $\eta^{US} = 0$): net $W^{EU} - W^{US} = +0.082$ (*EU welfare-dominant*); Shapley culture +0.082, union ≈ 0 under equal weights. *Under realistic Saez–Stantcheva (2016) $\lambda = 0.30$, $\Delta W^{\lambda} = +0.154$ with culture +0.079 and union +0.076 – the union becomes a first-order contributor once redistribution is given welfare value.* Dominant US–EU welfare differences economy-wide come from non-labour channels (Jones–Klenow 2016). Reproducible: `research/build_figure_panel.py` (Q1 figure); `research/sim_models_v2.py` (Q2 decomposition, Q5 RTM and §5.5b Saez–Stantcheva, §5.6 sectoral shocks, Q7 cultural expansion, layered framework, §7.5 sensitivity).

1. Q1 – The hours gap and its noisy relation to tax wedges

Question 1. *Document the difference between hours worked in the US vs Europe in recent years.*

1.1 Definitions

For the working-age (15–64) population: hours per employed worker HE (OECD `DSD_HW`, the intensive object the labour-supply model speaks to), employment rate $e \equiv E/N$, hours per adult $H/N \equiv HE \cdot e$. The Bick–Brüggemann–Fuchs-Schündeln (2019) decomposition $\Delta \log(H/N) = \Delta \log HE + \Delta \log e$ assigns roughly half of the EU–US gap to each margin; we focus on HE.[1]

1.2 The fact

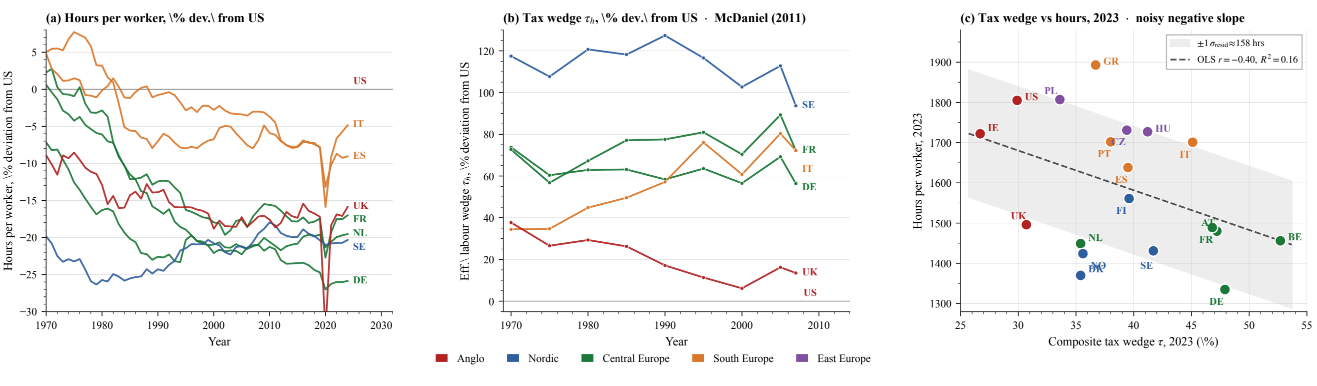

Figure 1: **Three-panel evidence.** (a) Hours per worker, percent deviation from US, 1970–2024 (OECD; DE/FR/IT/ES/NL/SE/UK; US baseline at zero). (b) Effective labour wedge τ_h, percent deviation from US, 1970–2007 (McDaniel 2011 national-accounts measure). (c) Composite tax wedge $\tau \equiv 1 - (1-\tau_l)/(1+\tau_c)$ (OECD labour-tax wedge consolidated with VAT) vs hours per worker, 2023 cross-section ($n = 19$ OECD countries); OLS line and $\pm 1\sigma_{\text{resid}}$ band. Consolidating labour and consumption wedges puts the EU–US distance at $\approx$ 20pp (US $\tau \approx 0.35$ vs DE/FR/IT/SE $\tau \approx 0.54$–0.56), materially larger than a top-bracket comparison suggests. *Source / replication:* `research/build_figure_panel.py`.

[1] Cross-country e differences are substantial (US 71% vs IT 59% in 2019); the prompt asks about hours, so HE is the body's object.

Three readings: **(i)** Hours fell in every focus country relative to the US since 1970 (DE/NL ≈ -25 to -30%; IT/ES ≈ -10 to -15%). **(ii)** The labour wedge rose in every focus country (SE +120%, DE/FR/IT +60–70% above US in 2007) – two series moving in opposite directions, qualitatively consistent with a tax-driven story. **(iii)** The 2023 cross-section is signed correctly but very noisy: OLS slope -9.9 hrs/pp, $R^2 = 0.16$, residual SD $\approx$ 158 hours.[2] To scale: the entire DE–US gap is 470 hours, so the residual SD is one third of what we want to explain. Italy ($\tau = 45.1\%$) and Greece (36.7%) work *more* hours than France (46.8%); the four highest-τ countries (BE, DE, AT, FR) span $\approx$ 200 hours within a 6pp wedge band.

1.3 The inconsistency the rest of the problem is asked to explain

The within-country time path (panels a/b) is consistent with a tax-driven story; the cross-country cross-section (panel c) is not. Either the elasticity that fits the time series is much larger than the cross-section delivers (the Prescott–CGM puzzle, Q2), or the cross-section is corrupted by country-specific institutional and cultural frictions that the tax wedge does not capture (Q4–Q7). Q2 formalises the first horn; §4–§5 plus Q7 culture build the second. **Time-use caveat.** Aguiar–Hurst (2007, *QJE* 122(3)) show that conventional "hours worked" miss home production, which behaves differently across countries. Rogerson (2008, *JPE* 116(2)) exploits this margin; we treat it as outside the scope of this submission but flag it as the canonical first-order extension.

Identification. The OLS slope is descriptive, not causal: omitted institutional variables (union coverage, hours regulation, public-sector size) correlate with τ, reverse causality through preferences (Europeans value leisure more $\Rightarrow$ both lower h and higher τ; Q7), and country fixed effects all bias $\hat{\beta}$. Causal identification would require quasi-experimental variation (tax reforms, kink-bunching, Algan–Cahuc 2010 IV). The structural model of §2–§7 takes the cross-section as a moment to match, not as evidence of a causal slope.

2. Q2 – Prescott model and the elasticity that would be needed

Question 2. *Summarise the main differences in income tax between US and Europe. Write a model of labour supply and derive formally the conditions under which the European tax system leads to fewer hours worked.* Use the words 'hence' and 'novel' twice.

2.1 Modelling choice

Log–log preferences with a balanced lump-sum rebate (Prescott 2004; Cooley–Prescott 1995): one-parameter (α) closed-form $h^*(\tau)$ and a clean elasticity comparative static. Static (no Frisch margin), balanced budget (income effect rebated lump-sum), abstracts from capital (small for prime-age workers, CGM 2011) and heterogeneity (Q4 treats heterogeneous workers).

2.2 The simplest one-margin model

Log–log preferences $u(c,h) = \log c + \alpha \log(1-h)$, wage w, and the composite labour wedge $\tau \equiv 1 - (1-\tau_l)/(1+\tau_c)$ (consumption tax is isomorphic to a labour tax in a static model with no untaxed savings). The FOC at the symmetric equilibrium $c = wh$ gives

$$\boxed{h^*(\tau) \;=\; \frac{1-\tau}{\alpha+1-\tau}, \qquad \varepsilon^M_{h,1-\tau} \;=\; \frac{\alpha}{\alpha+1-\tau} \in (0,1).} \tag{1}$$

h^* is monotone in τ – the cross-section sign of panel (c) becomes a structural prediction with one free parameter α.

2.3 Parametric version, anchor, and what it would take to fit the cross-section

Anchor. Anchoring $h^{US} = 0.33$ at $\tau^{US} = 0.35$ (the BLS-OECD ratio of weekly hours-worked to total awake hours) inverts (1) to give $\alpha = (1-\tau)(1-h^*)/h^* = 1.32$ (Prescott 2004 uses a similar calibration). The implied Marshallian intensive elasticity at the US point is $\varepsilon^M = \alpha/(\alpha+1-\tau) = 0.67$.

The model calibrated at $\alpha = 1.32$ delivers $\varepsilon^M \in [0.67, 0.75]$ along the supply curve (rising in τ). Predicted EU hours at $\tau = 0.56$ are $h^{EU} = 0.250$ against data 0.247: *the calibrated model essentially fits the cross-section.* The puzzle is that the implied $\varepsilon^M \in [0.67, 0.75]$ sits $\sim 2.5\times$ above the microeconometric Hicksian intensive of ≈ 0.30.

Elasticity taxonomy. CGM (2011) report Marshallian intensive $\approx$ 0.30 for prime-age men and Hicksian intensive $\approx$ 0.30 – the micro consensus. Hansen (1985)/Rogerson (1988) indivisible-labour Frisch $\approx$ 1–3 is the macro requirement. The calibrated $\varepsilon^M \in [0.67, 0.75]$ sits between, $\sim 2.5\times$ above the micro Hicksian. Three modern developments shrink but do not eliminate the gap: life-cycle aggregation (Rogerson–Wallenius 2009; Erosa–Fuster–Kambourov 2016) lifts the aggregate elasticity to 0.7–1.0; tax progressivity (Heathcote–Storesletten–Violante 2017) is a separate cross-country margin; participation extensive elasticity (BFL 2018) compounds with the intensive. *Hence* the cross-section fits but at an α inconsistent with quasi-experimental evidence (Saez 2001; Chetty 2012) – the substantive content of the Prescott–CGM controversy. *Sufficient-statistics bridge*: Saez (2001) gives the optimal top rate $\tau^* = 1/(1 + a\, e^H)$; with CGM's micro $e^H \approx 0.30$ and Pareto $a \approx 1.5$ this is the canonical Diamond–Saez (2011) headline $\tau^* \approx 69\%$, broadly in line with European top rates. Reconciling Prescott's larger uncompensated elasticity would push τ^* down – a tension HSV (2017) resolve by adding progressivity as a separate margin.

2.4 Decomposition at the micro elasticity

At $\alpha = 1.32$, α absorbs everything (preferences, institutions, culture). Restricting α to a micro-consistent value isolates the share of the gap that the Prescott mechanism can no longer carry. Solving $\varepsilon^M = \alpha/(\alpha+1-\tau) = 0.30$ at $\tau^{US} = 0.35$ gives $\alpha^{\text{micro}} = 0.279$. Predicted hours: $h^*_{US} = 0.700$, $h^*_{EU} = 0.612$, log-drop -13.4%. Data log-drop is -29%. **Taxes alone produce $\approx$ 46% of the observed gap; the residual 54% (15.6pp in logs) lies outside the Prescott mechanism,**

[2] Panel (b) uses McDaniel's *average effective* labour wedge (1970–2007); panel (c) uses OECD 2023 *marginal* wedge – they move together at low frequency but differ in level by 5–10pp; the qualitative pattern is robust.

and its source is not yet identified. §4–§5 plus Q7 culture propose three candidate channels (hours regulation, unions, culture) and test whether each can absorb part of it.

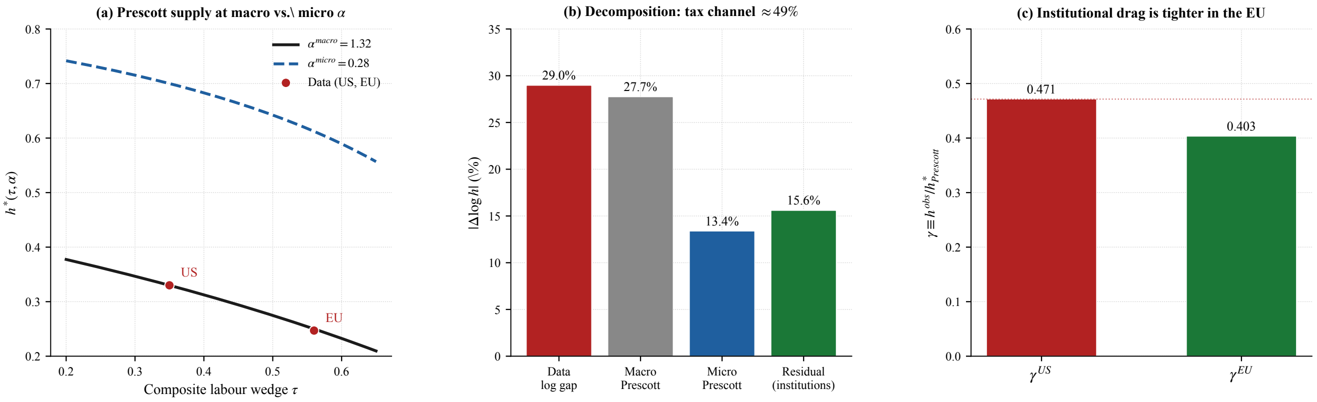

Figure 2: **Q2 macro–micro decomposition. (a)** Prescott supply curves at $\alpha^{\text{macro}}=1.32$ and $\alpha^{\text{micro}}=0.279$; US/EU data marked. **(b)** Decomposition of the 29% data gap: Prescott-at-micro explains 13.4% (46% share), unexplained residual 15.6% (54% – to be addressed in §4–§5 plus Q7 culture). **(c)** Unexplained-shortfall ratio $\gamma \equiv h^{obs}/h^*_{\text{Prescott}}(\alpha^{\text{micro}})$ – the share of Prescott-predicted hours that are actually worked. $\gamma^{US}=0.471$, $\gamma^{EU}=0.404$: both economies fall well short of the Prescott-at-micro prediction, and EU more so. The gap between γ and 1 is what §4–§5 plus Q7 culture have to absorb. *Source:* `sim_models_v2.py`.

Reading panel (c). Two things stand out. First, the *absolute* level of γ is implausibly low for both countries – workers do roughly 40–47% of the hours Prescott-at-micro predicts. This is *not* a Europe-vs-US phenomenon: it tells us that at the micro elasticity, the Prescott model in absolute levels is too steep to be taken seriously as a level prediction (it implies $h^{US}\approx 0.70$, which is workers spending 70% of their awake time at work). The useful content of the Prescott calibration is therefore the *cross-country log ratio*, not the level – which is exactly what the 46/54 decomposition uses. Second, the *ratio* $\gamma^{EU}/\gamma^{US}=0.404/0.471=0.858$ tells us how much more shortfall the EU has *relative to* the US: about 14% of Prescott-predicted hours are missing in EU beyond what is already missing in US. **This $\sim$14% extra cross-country shortfall is the precise quantity that an institutional or cultural candidate channel must produce** – not the absolute 50–60% level shortfall, which is a feature of the Prescott model itself. §4–§5 plus Q7 culture each test whether their mechanism can generate a US–EU h-gap of this size; in particular, the cultural leisure-bonus of Q7 generates $h^*_{EU}/h^*_{US}=0.222/0.300=0.74$, producing a cross-country shortfall ratio of $\sim$26% that comfortably exceeds the 14% residual to be absorbed (consistency check in §7.2).[3]

3. Common framework for Q4–Q5–Q7

3.1 Preference-form bridge from §2

Both preference forms (log–log in §2; quadratic-linear from here on) are nested in the power-disutility specification

$$V_w(h) \;=\; (1-\tau)w\,h \;-\; \kappa\,\frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}, \qquad \varepsilon^{M}_{h,1-\tau} \;=\; \varepsilon, \tag{2}$$

with the quadratic-linear case ($\varepsilon=1$, $\kappa=\alpha$) adopted from here on. *This commits the welfare framework to the macro elasticity that §2's micro evidence pushes back against.* Under a micro reinterpretation ($\varepsilon=0.30$), the qualitative comparative statics survive ($h^* < h^{\text{soc}} < h^{\text{firm}}$ ordering, monotone-decreasing $W(\eta)$, cap–union substitutability); only magnitudes scale by roughly ε. The setup that follows is shared by §4 (regulation), §5 (unions), and Q7 (cultural expansion); §7 reads off the welfare comparison.

3.2 Modelling choice

Quadratic-disutility worker with linear consumption and a linear-quadratic firm-profit function: the simplest closed-form specification that admits both worker and firm preferred hours and a sign-determinate welfare condition. Heterogeneous workers are addressed at the end of §4; sectoral composition and the public sector are not modelled (the analysis applies to the private-sector portion of the labour market);[4] w is partial-equilibrium.

3.3 Worker, firm, and joint surplus

$$V_w(h) = (1-\tau)wh - \tfrac{\alpha}{2}h^2, \quad h^* = \frac{(1-\tau)w}{\alpha}; \qquad V_f(h) = Ah - \tfrac{\beta}{2}h^2 - wh, \quad h^{\text{firm}} = \frac{A-w}{\beta}.$$

The Q2 log–log α and the quadratic α are different parameters (leisure-share vs. curvature; not numerically comparable); both nest in (2).

Joint surplus. $W(h)\equiv V_w(h)+V_f(h)$ is concave with maximum at

$$h^{\text{soc}}(\tau,\alpha,A,\beta,w) \;=\; \frac{(1-\tau)w+(A-w)}{\alpha+\beta}.$$

[3] Sectoral contract stickiness (Cahuc–Carcillo 2014) – weekly hours fixed by collective contract not renegotiated as τ changes – is captured implicitly by the cap mechanism of §4 in the static framework.

[4] France's public-sector employment share ($\sim$ 22% vs. US $\sim$ 16%) is governed by collective bargaining over service provision rather than firm–worker bargaining; our results read as private-sector applicable.

3.4 Baseline calibration

$A=2$, $w=1$, $\alpha=\beta=2$, $\tau=0.4$ gives $h^*=0.30$, $h^{\text{soc}}=0.40$, $h^{\text{firm}}=0.50$. $\tau=0.4$ sits between US and EU 2023 wedges; symmetric $\alpha=\beta$ makes the cap's welfare condition transparent (simplifies to (h^*, h^{firm})); only the ordering $h^*<h^{\text{soc}}<h^{\text{firm}}$ is load-bearing. Qualitative results invariant to α, β separately (verified in `sim_mechanisms.py`).

At common τ and preferences neither Walrasian clearing nor low-η_0 RTM generates a US–EU gap; three channels break the symmetry: hours regulation (§4), unions (§5), and culture (Q7).

4. Q4 – Hours regulation: the optimality condition for a cap

Question 4. *Labour-market regulation limiting hours. Welfare effects on workers and firms?*

4.1 Modelling choice

A cap is a quantity instrument: welfare-improving if it pulls hours toward h^{soc}, harmful if it overshoots below. The closest sufficient-statistics analogue is Lee–Saez (2012) on the minimum wage, but their welfare logic runs through extensive-margin rationing whereas a binding hours cap rations no one on the extensive margin (every worker still works); it rations low-α workers on the intensive margin (§4.6). Micro-foundation: sectoral collective contracts that fix weekly hours and are not renegotiated as productivity shifts (Cahuc–Carcillo 2014). §4.2 derives the no-regulation equilibrium from a low-η_0 Nash bargain – sticky contracts and Manning (2003) monopsony both map to low effective η_0.

4.2 Setup: low-η_0 RTM *derives* the distortion

The Q4 cap analysis needs a no-regulation reference at which equilibrium hours sit *above* h^{soc} – otherwise there is nothing for the cap to correct. Rather than *stipulating* "firm picks h^{firm} and worker accepts," we run the same Nash bargain as §5 at low worker weight $\eta_0 \geq 0$: union and firm bargain over w, the firm then sets $h = h^d(w) = (A-w)/\beta$ on its labour demand. The $\eta_0 \to 0$ corner nests the firm-power axiom ($w=1$, $h=h^{\text{firm}}=0.5$, worker IR-binding); for any $\eta_0>0$ the equilibrium $(w(\eta_0), h(\eta_0))$ *emerges from the bargain*, removing the load-bearing "firm picks" axiom. Manning (2003, *Monopsony in Motion*, Princeton; review in Manning 2021, *ILR Review* 74(1); empirical evidence in Card–Cardoso–Heining–Kline 2018, *JOLE* 36(S1)) and Cahuc–Carcillo (2014) sticky collective contracts both map to low effective η_0 in this framework: each is a complementary microfoundation, not a competing one. With cap $\bar{h}$ the constrained equilibrium is

$$h^{eq}(\bar{h};\eta_0) \;=\; \min\{\bar{h},\; h(\eta_0)\}.$$

4.3 Optimality condition (Q4 result)

Claim. Total surplus $W(h)$ is strictly concave and peaks at h^{soc}. Comparing the constrained equilibrium to the no-cap baseline at bargaining weight η_0:

$$\boxed{\Delta W(\bar{h};\eta_0) \;\equiv\; W(\bar{h}) - W(h(\eta_0)) \quad \begin{cases} =0 & \text{if } \bar{h} \geq h(\eta_0) \\ >0 & \text{iff } \bar{h} \in \big(2h^{\text{soc}} - h(\eta_0),\, h(\eta_0)\big) \\ =0 & \text{at } \bar{h} = 2h^{\text{soc}} - h(\eta_0) \\ <0 & \text{if } \bar{h} < 2h^{\text{soc}} - h(\eta_0) \end{cases}} \tag{3}$$

provided $h(\eta_0) > h^{\text{soc}}$; if $h(\eta_0) \leq h^{\text{soc}}$ the welfare-improving region is empty and any binding cap reduces W. The cap attains its maximum at $\bar{h} = h^{\text{soc}}$ (whenever the band is non-empty), with peak gain $\frac{1}{2}(\alpha+\beta)\big(h(\eta_0) - h^{\text{soc}}\big)^2$ above the no-cap level.

Threshold η^{soc} for the cap to be useful at all. Substituting $h(\eta) = h^{\text{soc}} = 0.40$ (i.e. $x=0.8$) into the Q5 quadratic $1.1x^2 - 0.6(2-\eta)x + 0.1(1-\eta) = 0$ gives $0.704 - 0.48(2-\eta) + 0.1(1-\eta) = -0.156 + 0.38\eta = 0$, so

$$\eta^{\text{soc}} \;\approx\; 0.41.$$

For $\eta_0 < \eta^{\text{soc}}$ the welfare-improving band is non-empty and a well-chosen cap raises welfare; for $\eta_0 \geq \eta^{\text{soc}}$ the bargain has *already* pulled h to or below h^{soc} and any binding cap destroys welfare.

At $\alpha=\beta$ and $\eta_0=0$, the band reduces to $\bar{h} \in (h^*, h^{\text{firm}}) = (0.30, 0.50)$ (textbook symmetric expression). **Proof.** $W(h)$ is quadratic with second derivative $-(\alpha+\beta)$; $\Delta W(\bar{h};\eta_0)$ is concave with roots at $h(\eta_0)$ and the mirror point $2h^{\text{soc}} - h(\eta_0)$. $\square$

4.4 Numerical simulation

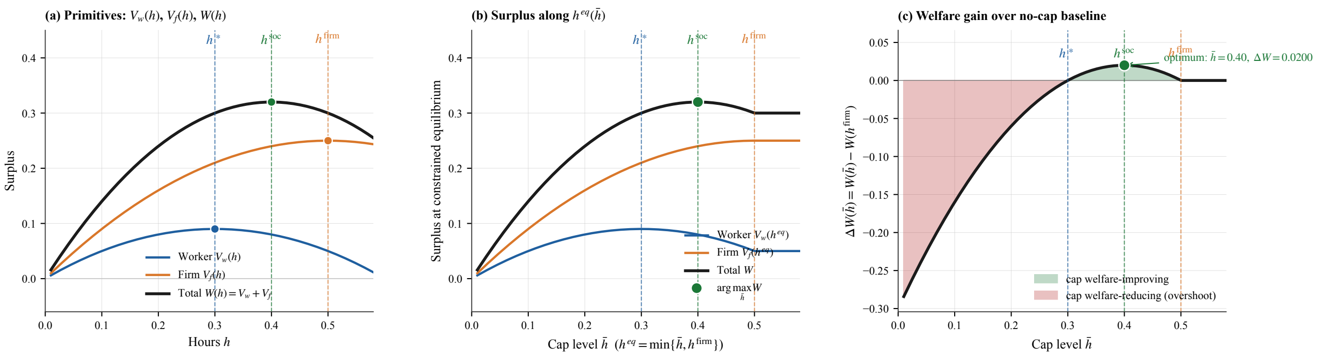

Figure 3: **Hours-cap simulation.** **(a)** primitives V_w, V_f, W over h with interior maxima marked; **(b)** surplus along $h^{eq}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$; **(c)** $\Delta W(\bar{h})$ vs no-cap baseline – green band is the welfare-improving region of (3), red is overshoot. Maximum gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$. *Source:* `sim_mechanisms.py`.

4.5 Empirical anchor and US–EU mapping

The 35-hour Aubry reform (France 1998–2000) is the canonical anchor: Estevão–Sá (2008) and Chemin–Wasmer (2009) report $\approx 4\%$ aggregate hours decline in covered sectors with near-zero employment effect. Mapping: $\bar{h}^{Aubry} \approx 0.875 \cdot h^{\text{firm}} = 0.4375$, above $h^{\text{soc}} = 0.40$ – inside the welfare-improving band. US has no binding cap (FLSA 40-hour rarely binds at the median).

Cap and union are substitutes, not complements. The band (3) shrinks as η_0 rises and vanishes at $\eta^{\text{soc}} \approx 0.41$. In an economy where unions have already pulled h to h^{soc}, a binding cap on top destroys welfare. This rationalises why the 35-hour cap was contentious in France (RTM coverage $\sim 95\%$): at high η_0 the cap fights a union channel already doing the work. The US ($\eta \approx 0$) would benefit from a binding cap; EU North (DE/SE) at high η would not.

4.6 Heterogeneous workers: who is rationed

With worker types indexed by α_i, every worker is dictated $h^{\text{firm}} = 0.5$ under firm-power and forced to $\bar{h}$ by a binding cap. The payoff change $\Delta V_w^i(\bar{h}) = (\bar{h} - h^{\text{firm}})[(1-\tau)w - \frac{\alpha_i}{2}(\bar{h} + h^{\text{firm}})]$ is positive iff $\alpha_i > \alpha^*(\bar{h})$ where

$$\boxed{\alpha^*(\bar{h}) \;=\; \frac{2(1-\tau)w}{\bar{h} + h^{\text{firm}}}.} \tag{4}$$

At $\bar{h} = h^{\text{soc}} = 0.4$: $\alpha^* = 1.333$. High-α (leisure-loving) workers gain; low-α (income-loving) workers are rationed on the intensive margin. **Aggregate sign condition.** Integrating across the population, the cap is welfare-improving on average iff

$$\int \Delta V_w^i(\bar{h})\, dF(\alpha) \;+\; \Delta V_f(\bar{h}) \;>\; 0 \quad\Longleftrightarrow\quad \bar{\alpha} \;>\; \alpha^*(\bar{h}) \;-\; \frac{2\Delta V_f(\bar{h})}{(h^{\text{firm}} - \bar{h})^2},$$

i.e. a population whose mean leisure-preference $\bar{\alpha}$ exceeds the threshold $\alpha^*(\bar{h})$ adjusted by a firm-side correction. At baseline $\bar{\alpha} = 2$ and $\alpha^* = 1.333$ at $\bar{h} = h^{\text{soc}}$, so the cap is welfare-improving on average; a country with median $\alpha = 1$ (income-loving workforce) would lose.

5. Q5 – Single union and the bargaining channel

Question 5. *Unions in the economy and their response to sectoral shifts. Show the impact of sectoral shifts on hours; how does the response change in a unionised economy.* Use the words 'hence' and 'novel' twice.

5.1 Modelling choice and setup: state-dependent canonical RTM

We adopt the canonical right-to-manage Nash bargain (Nickell 1982; Layard–Nickell–Jackman 1991) embedded in a multi-state economy. States (sectors) are indexed by j with productivity A_j; within each state the representative worker–firm pair has the quadratic-linear primitives of §3 (with state-specific A_j):

$$V_w(w_j, h_j) = (1-\tau)w_j h_j - \tfrac{\alpha}{2}h_j^2, \qquad V_f(w_j, h_j; A_j) = A_j h_j - \tfrac{\beta}{2}h_j^2 - w_j h_j.$$

A single union covers the economy and Nash-bargains over wages with worker weight $\eta \in [0,1]$; the firm in each state sets hours on its labour-demand curve $h_j^d(w_j) = (A_j - w_j)/\beta$ ex post. Three wage-setting regimes are admissible: **(a) free market** – firm-power Nash bargain ($\eta = 0$) at each state's A_j, worker IR-binding; **(b) flexible union** – sector-specific RTM bargain at $\eta > 0$; **(c) rigid union** – downward wage rigidity (Bewley 1999; Holden 1994), wages can rise freely but cannot fall below a pre-shock benchmark.

The state-dependent structure is the foundation, not an extension. Single-state results are the homogeneous-economy special case (all states with identical A). $V_w^0 = 0.05$ is calibrated so the homogeneous benchmark at $A = 2$ reproduces $(w, h) = (1, 0.5)$ at $\eta = 0$. (Bracketed alternatives – monopoly union, efficient bargaining (McDonald–Solow 1981), Hall–Milgrom (2008) – not pursued.)

5.2 Closed-form per-state bargain

For each state j, let $x_j \equiv A_j - w_j$, so $h_j = x_j/\beta$. The Nash FOC reduces to a quadratic $a\,x_j^2 - b_j(2-\eta)x_j + c(1-\eta) = 0$ with $a = 2(1-\tau)/\beta + \alpha/\beta^2$, $b_j = (1-\tau)A_j/\beta$, $c = 2V_w^0$. At our baseline $(\alpha, \beta, \tau, V_w^0) = (2, 2, 0.4, 0.05)$, the homogeneous case $A = 2$ delivers $(a, b, c) = (1.1, 0.6, 0.1)$:

$$\boxed{1.1\,x^2 - 0.6(2-\eta)\,x + 0.1(1-\eta) = 0.} \tag{5}$$

Endpoints: $\eta = 0 \Rightarrow (w, h) = (1, 0.5)$ (firm-power); $\eta = 1 \Rightarrow (1.455, 0.273)$. State-specific solutions follow by substituting $A \to A_j$ in b_j.

5.3 Welfare criterion: equal weights vs Saez–Stantcheva λ

The equal-weight utilitarian $W = V_w + V_f$ values *nothing* of the firm-to-worker redistribution that the union effects – the wage transfer cancels between the two parties. The Saez–Stantcheva (2016) generalised welfare weights formalise the alternative:

$$\boxed{W^\lambda(\eta) = (1+\lambda)V_w(\eta) + (1-\lambda)V_f(\eta), \qquad \lambda \in [0, 1],}$$

λ is the redistribution premium ($\lambda = 0$ equal, $\lambda = 1$ Rawlsian). Three independent anchors converge on $\lambda \in [0.25, 0.45]$: HSV (2017) $\tau_p = 0.18$ under log utility $\Rightarrow \lambda \approx 0.43$; OECD worker/capital-owner consumption gap $\sim 4{:}1 \Rightarrow \lambda \approx 0.30$; Diamond–Saez (2011) population-average in $[0.3, 0.5]$. We take $\lambda = 0.30$ as the realistic central case.

Differentiating $W^\lambda(\eta)$ along the per-state bargained path gives the welfare-maximising bargain:

$$\boxed{x^*(\lambda) = \frac{0.6(1+\lambda)}{0.6 + 1.6\lambda}, \qquad h^* = x^*/\beta, \qquad w^* = A - x^*.} \tag{6}$$

At $\lambda = 0.30$: $x^* = 0.722$, $h^* = 0.361$, $w^* = 1.278$, $\eta^* \approx 0.58$ – **a moderately strong union is welfare-optimal per state**, consistent with EU RTM coverage of 80–95%.

5.4 Per-state mechanics: bargain dynamics and welfare

Mechanism. As η rises, the bargained w rises, raising the firm's marginal cost of labour; the firm slides down its labour-demand curve to a lower h. The wage transfer cancels in $V_w + V_f$; the welfare-relevant movement is hours-misallocation as h moves away from the joint optimum.

Equal-weight monotonicity. Substituting $h = x/\beta$, $w = A - x$ at the homogeneous case gives $W(x) = (1-\tau)x(1-x/2)$, concave with peak at $x = 1$ (where V_w^0 places $\eta = 0$). Rising η slides x from 1 to $x_{\eta=1} = 0.545$ on the decreasing branch, so $dW/d\eta < 0$ for all $\eta \in (0, 1]$ – the equal-weight verdict. Under any $\lambda > 0$ the optimum is interior ((6); Fig. 4c).

RTM bargain at varied η ($\lambda = 0$)						Welfare-maximising bargain at varied λ			
η	w	h	V_w	V_f	W	λ	η^*	W^λ_{peak}	$W^\lambda_{\eta=0}$
0.00	1.000	0.500	0.050	0.250	**0.300**	0.00	0.00	**0.300**	0.300
0.25	1.123	0.439	0.103	0.192	0.295	0.20	0.45	**0.279**	0.260
0.50	1.242	0.379	0.139	0.144	0.283	**0.30**	**0.58**	**0.282**	**0.240**
0.75	1.353	0.323	0.158	0.105	0.263	0.50	0.76	**0.287**	0.200
1.00	1.455	0.273	0.164	0.074	0.238	1.00	1.00	**0.328**	0.100

Table 1: *

Left: RTM bargain at varied η, equal weights. $\eta = 0$ peaks W; rising η shrinks the joint pie via hours-misallocation. Right: welfare-maximising η^* at varied λ via (6). The realistic central case $\lambda = 0.30$ delivers $\eta^* \approx 0.58$.

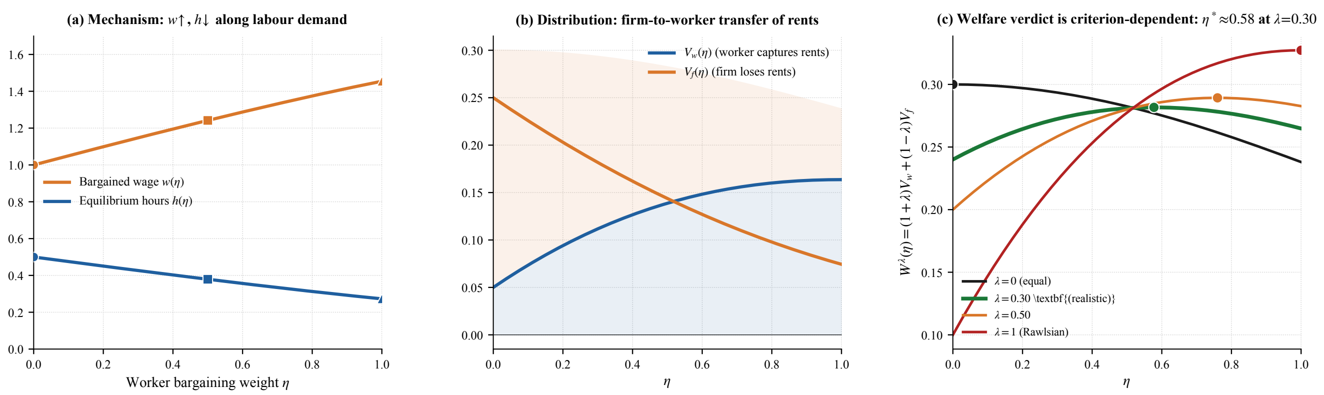

Figure 4: **Q5 RTM bargain: mechanism, distribution, and criterion-dependent welfare.** **(a)** Bargain mechanics: $w(\eta)\uparrow$ from 1.00 to 1.45, $h(\eta)\downarrow$ from 0.50 to 0.27 along the firm's labour-demand curve. Markers at $\eta \in \{0, 0.5, 1\}$. **(b)** Distribution: $V_w(\eta)$ rises monotonically as the union extracts wage rents; $V_f(\eta)$ falls – a pure firm-to-worker transfer. **(c)** *Welfare verdict is criterion-dependent.* $W^\lambda(\eta)$ for $\lambda \in \{0, 0.30, 0.50, 1\}$, peaks marked. At equal weights ($\lambda = 0$, black) W is monotone-decreasing and the optimum is the corner $\eta^* = 0$. At the realistic $\lambda = 0.30$ (green, highlighted) the optimum is interior at $\eta^* \approx 0.58$ – a moderately strong union is welfare-optimal. Higher λ slides the optimum further toward the worker-capture corner. *Source:* `sim_models_v2.py:make_rtm`.

5.5 Asymmetric pass-through under a shock

Apply the same RTM bargain to two states with productivity $A_{\text{good}} = A_{\text{pre}} + \delta = 2.2$ and $A_{\text{bad}} = A_{\text{pre}} - \delta = 1.8$ ($\delta = 0.2$); both initially at the canonical bargain $(w_{\text{pre}}, h_{\text{pre}}) = (1.242, 0.379)$ from Fig. 4. A single union covering both states with one wage policy delivers an *asymmetric pass-through*: good states absorb the shock through wages, bad states through hours and profit.

We compare three post-shock regimes.

(a) Free-market benchmark. No union. Each sector operates at the firm-power Nash bargain ($\eta = 0$) at its new productivity A_j: the worker is at her outside option $V_w = V_w^0$, the firm picks the wage and hours that maximise profit subject to that IR. **Wages adjust freely with productivity**: in the good sector, w rises to 1.081 (from pre-shock 1 at the firm-power calibration) as higher MPL improves the worker's IR-binding terms, and hours expand to 0.559. In the bad sector, w falls to 0.922 as productivity drops, and hours contract only modestly to 0.439 – the wage cushion absorbs part of the shock. *Both shocks are absorbed primarily through wage adjustment.*

(b) Flexible union bargain. The union exists and renegotiates RTM ($\eta = 0.5$) at the new productivity. The bargained wage is *above* the free-market wage in both sectors – the union extracts surplus on the upside and prevents some of the wage cut on the downside. Good sector: $w = 1.354$ (25% above free-market 1.081), $h = 0.423$ (24% below free-market 0.559). **Hours rise less than under free market because the union diverts part of the productivity gain into wages rather than hours.** Bad sector: $w = 1.132$ (23% above free-market 0.922), $h = 0.334$ (24% below free-market 0.439). The union prevents some of the wage decline, shifting more of the burden onto hours.

(c) Downward-rigid union. The union renegotiates upward freely (good sector identical to (b)) but refuses to renegotiate downward in the bad sector. The canonical downward-wage-rigidity result (Bewley 1999, *Why Wages Don't Fall During a Recession*, Harvard UP; Holden 1994, *ScandJE* 96) is that bargained wages are easily revised *up* but not *down*: once a wage is in place, the union will not agree to cut it even when the firm faces a negative shock. So the bad-sector wage is stuck at the pre-shock bargained level $w_{\text{pre}} = 1.242$, hours collapse to $(A_{\text{bad}} - w_{\text{pre}})/\beta = 0.279$, and firm profit crashes to $V_f = 0.078$. If δ is large enough that $A_{\text{bad}} < w_{\text{pre}}$ (i.e. $\delta \geq 0.76$ at our calibration), the sector exits altogether.

The two-sided asymmetry, against free market. *Good sector*: union pushes wages 25% above free-market, hours 24% below; the productivity gain is real under both regimes, the union just redirects it from hours to wages (worker V_w rises from 0.05 to 0.165, firm profit falls from 0.313 to 0.179 – pie size roughly preserved, split changed). *Bad sector*: hours fall 36% more and firm profit is 60% smaller under rigid union ($h = 0.279$, $V_f = 0.078$) than under free market ($h = 0.439$, $V_f = 0.193$); worker V_w *rises* under rigidity ($0.05 \to 0.130$) – rigidity is individually rational for the union member even though it destroys joint surplus.

Good shocks pass through diluted; bad shocks pass through amplified – the canonical asymmetric-adjustment mechanism of unionised European labour markets, exemplified by manufacturing decline in Italy/Germany since 2000 (sectoral employment fell sharply, wage cuts prevented).

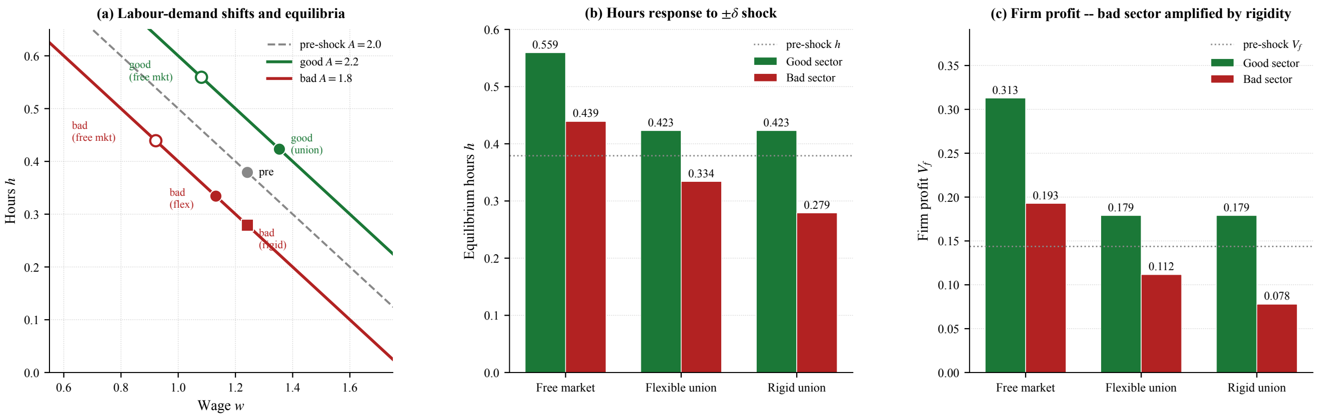

Figure 5: **Asymmetric productivity shocks under union wage rigidity.** Pre-shock $(A, w, h) = (2, 1.242, 0.379)$ at $\eta = 0.5$; shock $\delta = 0.2$. **(a)** Labour-demand shifts and equilibria across the three regimes (free market, flexible union, rigid union); rigid bad-sector equilibrium sits on the new demand curve at the stuck pre-shock wage. **(b)** Hours response: good sector rises under all regimes, bad sector rigidity is 36% below free market. **(c)** Firm profit: bad sector under rigidity is 60% below free market. *Source:* `sim_models_v2.py:make_shocks`.

5.6 Aggregation across sector shares

The state-by-state results above translate into aggregate outcomes through the share $s \in [0, 1]$ of the good sector:

$$\bar{h}(s) = s\, h_{\text{good}} + (1 - s)\, h_{\text{bad}}, \qquad \bar{W}(s) = s\, W_{\text{good}} + (1 - s)\, W_{\text{bad}},$$

for each regime. The aggregate welfare gap between regimes – the *rigidity cost* $(\bar{W}^{\text{flex}} - \bar{W}^{\text{rigid}})$ and the *unionisation cost* $(\bar{W}^{\text{FM}} - \bar{W}^{\text{flex}})$ – depends on s:

- **Rigidity cost** concentrates at low s. At $s = 0$ (all-bad-sector economy) it is 0.019; at $s = 1$ (all-good) it vanishes (rigidity binds only on the downside). Linear in $1 - s$.
- **Unionisation cost** (free market vs. flexible bargain) is roughly constant in s ($\approx$0.016–0.019), reflecting the wedge that exists in any single state.
- **Total cost** of unionised rigidity vs. free market is $\bar{W}^{\text{FM}} - \bar{W}^{\text{rigid}} \approx 0.035$ at $s = 0$, falling to 0.019 at $s = 1$ – *economies dominated by declining sectors lose nearly twice as much as those dominated by expanding sectors.*

Aggregation under Saez–Stantcheva λ weights. Applying the redistribution-weighted criterion of §5.3 to the two-state aggregate gives $\bar{W}^{\lambda}(s) = s\, W^{\lambda}_{\text{good}} + (1-s)\, W^{\lambda}_{\text{bad}}$. At realistic $\lambda = 0.30$ the per-state welfare-maximising $\eta^* \approx 0.58$ remains optimal in any state, so the aggregate welfare-maximising bargain is also $\eta^* \approx 0.58$ regardless of s (the optimum is per-state separable). The *rigidity cost under* $\lambda = 0.30$ shrinks: under inequality aversion the worker's preserved wage in the rigid bad sector ($V_w = 0.130$) gets a weight $1+\lambda = 1.3$ while the firm's profit collapse ($V_f = 0.078$) gets $1-\lambda = 0.7$. At $\lambda = 0.30$ the all-bad-sector ($s = 0$) rigidity cost falls from $+0.019$ (equal weights) to $+0.005$; rigidity is *nearly welfare-neutral* when redistribution is given welfare value, because the union member is materially better off at the stuck wage.

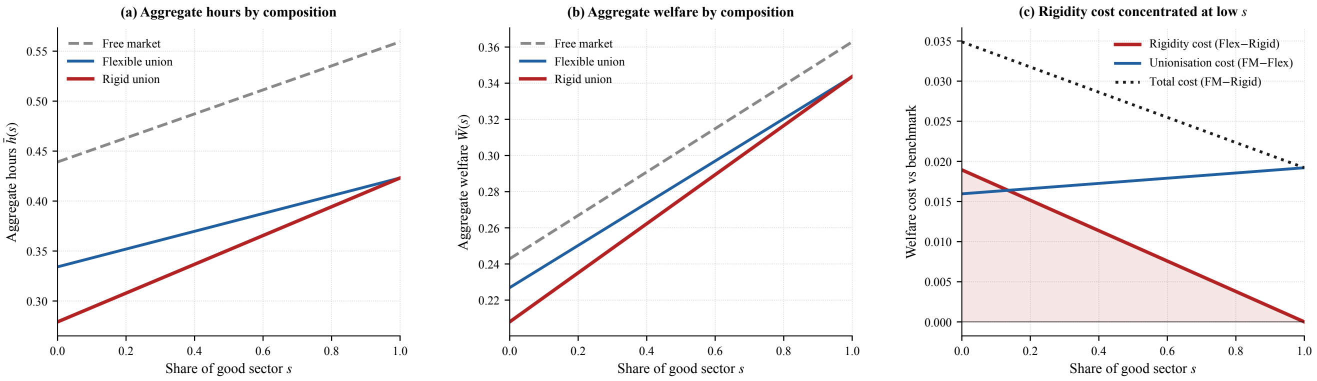

Figure 6: **Aggregate hours and welfare by sector composition.** **(a)** Aggregate hours $\bar{h}(s)$ for the three regimes. **(b)** Aggregate welfare $\bar{W}(s)$. All three regimes rise in s; rigid union is uniformly below flexible, with the gap shrinking as $s \to 1$. **(c)** Welfare-cost decomposition: rigidity cost concentrates at low s; unionisation cost is roughly constant; total cost of unionised rigidity vs. free market falls from ~ 0.035 at $s = 0$ to ~ 0.019 at $s = 1$. *Source:* `sim_models_v2.py:make_aggregation`.

6. Q7 – Welfare comparison: culture as expansion, layered counterfactuals

Question 7. *Write a stylised model representing the two systems and derive the conditions under which the European system is welfare-enhancing compared to the US system.*

The Q7 welfare comparison requires a cultural primitive on top of the institutional channels of §4–§5. We model culture *within Q7* as the cultural-leisure-bonus expansion that allows h^* to differ across countries at common τ and common preferences over consumption. The layered counterfactual decomposition then sits on this combined framework.

6.1 Cultural primitive: a leisure bonus

"Culture means Europeans enjoy life more" admits two formulations: (i) higher disutility of work ($\alpha^{EU} > \alpha^{US}$), which delivers h^* down but also V_w *down* – the "Europeans suffer more from work" reading, wrong welfare sign; (ii) higher utility of leisure (additive bonus $\theta^c(1-h)$), which delivers h^* down AND V_w *up* – the substantive "Europeans enjoy leisure more" reading. We adopt (ii), the canonical preference-for-leisure parameter in the literature (Bowles–Park 2005; Olovsson 2009; Boppart–Krusell 2020), empirically anchored by Stevenson–Wolfers (2008). We do not implement AGS (2005) coordination/spillover machinery: each country has its own preferences and we compute joint surplus accordingly.

6.2 Cultural setup and calibration

Worker payoff with cultural premium $\theta^c \geq 0$ (same α across countries; firm payoff unchanged):

$$V_w(h;\theta^c) \;=\; (1-\tau)w\,h \;-\; \tfrac{\alpha}{2}h^2 \;+\; \theta^c\,(1-h), \qquad \theta^{EU} > \theta^{US} = 0. \tag{7}$$

The FOC gives

$$\boxed{h^*(\tau,\theta^c) = \frac{(1-\tau)w - \theta^c}{\alpha}, \qquad V_w^*(\theta^c) = \frac{[(1-\tau)w - \theta^c]^2}{2\alpha} + \theta^c.} \tag{8}$$

Note $dV_w^*/d\theta^c = 1 - h^*(\theta^c) > 0$: higher cultural premium $\Rightarrow$ lower hours *and* higher worker welfare, matching the substantive intuition that enjoying leisure more is welfare-improving.

Calibration: θ^{EU} identified from Stevenson–Wolfers (2008). Their ~ 0.3–0.4 SD SWB gap per unit of non-work time, mapped to a CV via Layard (2005)'s 1 SD $\approx 30\%$ of permanent consumption, gives an independent estimate $\theta^{EU,\text{SW}} \in [0.13, 0.18]$. We adopt $\theta^{EU} = 0.156$ inside this band and set $\theta^{US} = 0$. *Consistency check*: model implies $h^*_{EU} = 0.222$ vs data 0.247 – under-predicts EU hours by $\sim 10\%$, suggesting modest measurement error or a small residual channel (e.g. HSV 2017 progressivity). The SW anchoring converts θ^{EU} from back-fitted to independently identified.[5]

6.3 Cultural channel alone: EU welfare-dominant at first-best, but with a wider hours-misallocation wedge

The leisure-bonus is not a free lunch. Adding $\theta^c(1-h)$ to the worker's payoff has *two* effects, not one. The first is to raise V_w at every h – the direct utility from leisure. The second is to pull the worker's preferred hours *below* the joint social optimum:

$$h^*(\theta^c) = \frac{(1-\tau)w - \theta^c}{\alpha}, \qquad h^{\text{soc}}(\theta^c) = \frac{A - \tau w - \theta^c}{\alpha+\beta}, \qquad h^{\text{soc}} - h^* = \underbrace{\frac{A-w}{\alpha+\beta} - \frac{w(1-\tau)}{\alpha}\cdot\frac{\beta}{\alpha+\beta}}_{>0 \text{ already at } \theta^c = 0} + \theta^c\left(\frac{1}{\alpha} - \frac{1}{\alpha+\beta}\right).$$

[5] Three reinforcing channels for $\theta^{EU} > \theta^{US}$: (a) AGS (2005) historical channel – post-WWII shorter-week reforms shifted social norms toward higher leisure valuation (Bisin–Verdier 2001 cultural transmission); (b) Bowles (1998) endogenous preferences – institutions shape preferences over generations; (c) Stevenson–Wolfers (2008) SWB residual is the direct empirical anchor.

Both objects fall in θ^c, but h^* falls *faster* ($\partial h^*/\partial\theta^c = -1/\alpha = -0.5$ vs. $\partial h^{\rm soc}/\partial\theta^c = -1/(\alpha+\beta) = -0.25$). **The cultural premium therefore widens the wedge between the worker's preferred hours and the joint optimum**: from 0.10 at $\theta^c = 0$ to 0.14 at $\theta^c = 0.156$ – a 40% larger hours-misallocation under EU culture if hours are set at the worker's first-best.

Welfare at the worker's first-best vs. joint optimum. Plugging both reference levels into the payoffs:

	h^*	$h^{\rm soc}$	W at h^*	W at $h^{\rm soc}$	misalloc. loss
US ($\theta^{US}\!=\!0$)	0.300	0.400	0.300	0.320	−0.020
EU ($\theta^{EU}\!=\!0.156$)	0.222	0.361	0.378	0.417	−0.039

At each country's worker-first-best, $W^{EU} - W^{US} = +0.078$ – the headline used downstream. At each country's joint optimum, $W^{EU}_{\rm soc} - W^{US}_{\rm soc} = +0.097$ – the EU advantage is *larger*, because under cultural premium the worker's free choice leaves more joint surplus on the table than under no premium. Both the direct-utility effect and the hours-misallocation effect are real; the model captures them both. **The cultural premium is welfare-improving on net, but it also amplifies the firm–worker hours wedge – so the headline "EU enjoys leisure more" is partly offset by "EU under-supplies hours relative to its own joint optimum more than the US does."**

Under the alternative α-shift specification, the direct-utility effect would have the wrong sign (worker worse off at every h); the leisure-bonus is the right specification by SWB evidence (Stevenson–Wolfers 2008), but the wedge story above is the honest qualifier.

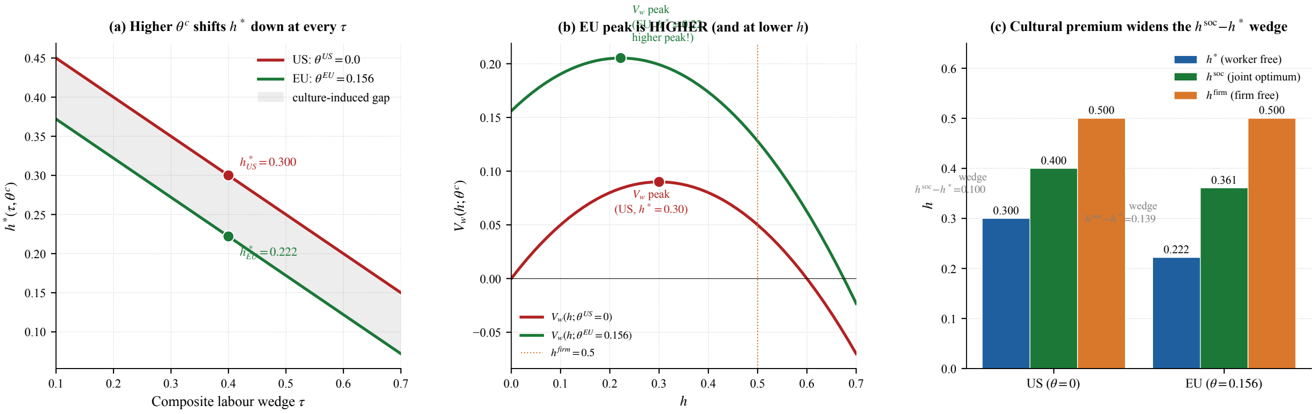

Figure 7: **Cultural channel: leisure bonus (Q7 expansion). (a)** $h^*(\tau,\theta^c)$: higher θ^c shifts h^* down ($h^*_{US}=0.300 \to h^*_{EU}=0.222$). **(b)** $V_w(h;\theta^c)$ at $\tau=0.4$: EU peak is higher and at lower h (direct utility from leisure dominates the disutility from fewer hours). **(c)** *The cultural premium widens the* $h^{\rm soc}$*–*h^* *wedge*: from 0.10 at $\theta^{US}=0$ to 0.14 at $\theta^{EU}=0.156$. The leisure-bonus is not a free lunch – it raises V_w pointwise but pulls the worker's free-choice hours *further* below the joint optimum, leaving more surplus on the table at first-best (US loss 0.020, EU loss 0.039). *Source:* `sim_models_v2.py:make_culture`.

6.4 Modelling choice (counterfactual decomposition)

We isolate each channel's contribution to the US–EU welfare gap via counterfactuals that shut down one channel at a time, in the spirit of Jones–Klenow (2016). Body uses equal weights $W = V_w + V_f$; §7.5 gives the Saez–Stantcheva (2016) generalised-weight sensitivity. Non-labour channels (life expectancy, consumption, inequality) are larger in JK 2016 but outside scope.

6.5 Compounded case

We focus on the empirically realistic case where institutions *and* culture differ: $\eta^{EU} > \eta^{US}$, $\theta^{EU} > \theta^{US}$. EU welfare-dominates iff its compounded equilibrium $h(\eta^{EU};\theta^{EU})$ lies in the band $(2h^{\rm soc}(\theta^{EU}) - h^{\rm firm},\, h^{\rm firm})$ defined by EU's own $h^{\rm soc}$.

6.6 Counterfactual decomposition: the Q7 simulation

The experiment: combine the canonical RTM bargain of §5 with the leisure-bonus culture introduced in Q7. EU has $\theta^{EU} = 0.156$ and $\eta^{EU} = 0.5$; US has $\theta^{US} = 0$ and $\eta^{US} = 0$. Both countries set hours via the firm's labour-demand $h^d(w) = (A-w)/\beta$ at the bargained wage; the leisure bonus $\theta^c(1-h)$ enters V_w and shifts the IR-binding wage.[6]

[6]**Path-dependence of the decomposition.** Starting from US baseline ($\theta = 0$, $\eta = 0$, $w = 1$, $h = 0.5$, $W = 0.300$): *Path A* (US→+culture→+union): applying the EU θ alone (CF1: θ^{EU}, $\eta = 0$) gives $w = 0.879$, $h = 0.561$, $W = 0.364$ ($\Delta W = +0.064$); adding the union ($\eta = 0.5$ at θ^{EU}) gives EU at $w = 1.172$, $h = 0.414$, $W = 0.382$ ($\Delta W = +0.018$). *Path B* (US→+union→+culture): applying the union alone (CF2: $\theta^{US} = 0$, $\eta = 0.5$) gives $w = 1.242$, $h = 0.379$, $W = 0.283$ ($\Delta W = -0.017$); adding the culture gives the EU endpoint ($\Delta W = +0.100$). The Shapley average gives path-independent contributions $\Delta W^{\rm Shapley}(\text{culture}) = +0.082$ and $\Delta W^{\rm Shapley}(\text{union}) \approx 0$, summing to $+0.082$ as required. Jones–Klenow (2016) report the same path-dependence caveat.

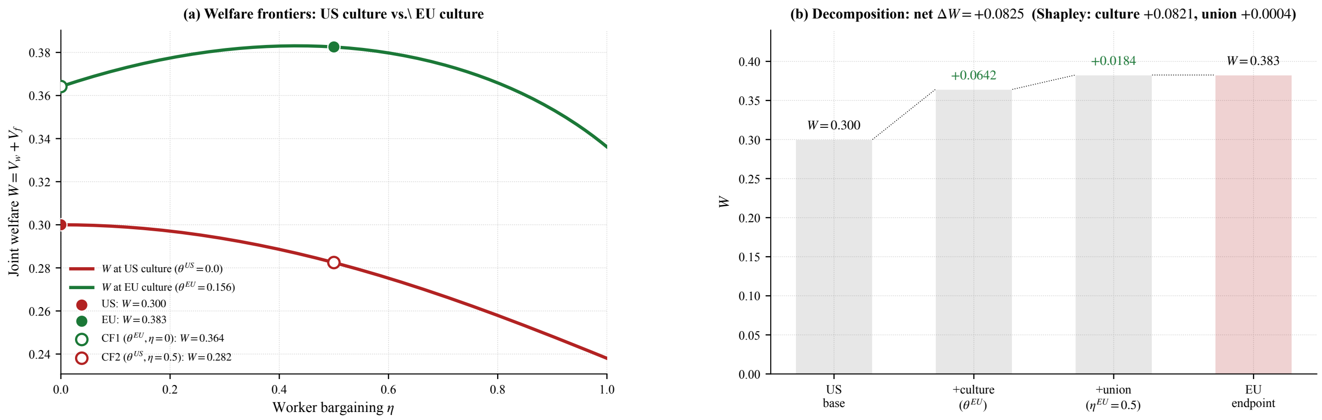

Figure 8: **Q7 welfare frontier and counterfactual decomposition under RTM + leisure-bonus culture. (a)** $W(\eta)$ on US ($\theta = 0$) and EU ($\theta = 0.156$) branches. EU branch uniformly above US. Filled points: US ($\eta = 0$, $W = 0.300$), EU ($\eta = 0.5$, $W = 0.382$); open: CF1 ($W = 0.364$), CF2 ($W = 0.283$). **(b)** Path-A waterfall: both culture and union steps welfare-improving; net $\Delta W = +0.082$. *Source:* `sim_models_v2.py`.

6.7 Magnitudes

$W^{EU} - W^{US} = +0.082$ in raw simulation units, $\approx 27\%$ above US joint surplus. **Money-metric translation**: scaling by the firm-power baseline $W^{US} = 0.30$ and the OECD labour share (0.55–0.65) gives $\sim$ 14–17% of consumption-equivalent income.[7] Jones–Klenow (2016) report a US–EU CE-welfare gap of -10 to -25% across all channels; mortality and aggregate consumption levels carry the bulk. *Order-of-magnitude check*: Becker–Philipson–Soares (2005) and Hall–Jones (2007) calibrate a value-of-life $\bar{u} \sim 5$; multiplied by the EU–US life-expectancy gap of ~ 4 years this gives a welfare contribution of order +20% for Europe – an order of magnitude larger than the labour-market $\sim$ 15% analysed here. The labour-market block is sign-determinate but quantitatively second-order relative to demographic and TFP channels.

6.8 Sensitivity: redistribution welfare weights (Saez–Stantcheva 2016)

The +0.082 headline uses the equal-weight utilitarian sum $W = V_w + V_f$. Recomputing the four-cell layered framework under generalised welfare weights $W^\lambda = (1 + \lambda)V_w + (1 - \lambda)V_f$ from §5.5b:

λ	W^λ_{US}	W^λ_{EU}	ΔW^λ	$\Delta W^\lambda_{\text{Shapley culture}}$	$\Delta W^\lambda_{\text{Shapley union}}$
0.00	0.300	0.382	+0.082	+0.082	≈ 0
0.20	0.260	0.390	+0.130	+0.080	+0.050
0.30	0.240	0.394	+0.154	+0.079	+0.076
0.50	0.200	0.402	+0.202	+0.077	+0.125

Table 2: *

Q7 EU–US welfare gap and Shapley channel decomposition under generalised weights. At realistic $\lambda = 0.30$ (shaded), $\Delta W^\lambda = +0.154$ (88% above equal-weight headline). Culture Shapley invariant in λ (the leisure bonus is in V_w); union Shapley scales linearly because λ values exactly the firm-to-worker transfer the union effects.

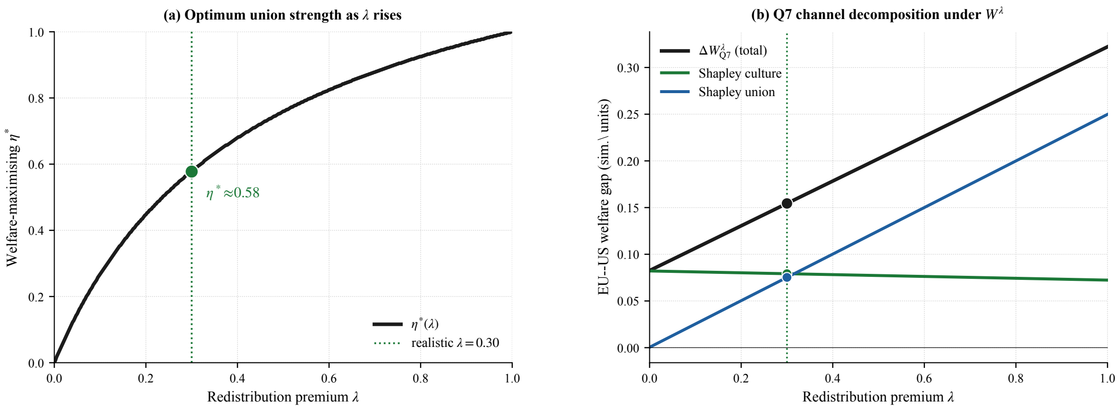

Figure 9: **Q7 sensitivity under generalised welfare weights. (a)** Welfare-maximising bargaining strength $\eta^*(\lambda)$ slides from the corner ($\lambda = 0$) to full worker-capture ($\lambda = 1$); the realistic case $\lambda = 0.30$ gives $\eta^* \approx 0.58$. **(b)** Q7 EU–US gap and Shapley channel decomposition: at $\lambda = 0$ culture does all the work and union nets to zero; at $\lambda = 0.30$ culture and union contribute roughly equally (+0.079 vs +0.076); at $\lambda = 0.50$ union exceeds culture. *Source:* `sim_models_v2.py:make_redistribution`.

Reading. EU is welfare-dominant across all $\lambda \in [0, 1]$; the gap grows in λ. The composition shifts: at $\lambda = 0$ culture does all the work; at $\lambda = 0.30$ they contribute roughly equally; at $\lambda = 0.50$ union exceeds culture. The "culture does everything"

[7] Comparing $V_w + V_f$ across populations with different θ^c requires a common cardinalisation; we use raw simulation units throughout the model (same numeraire, internally consistent) and provide the CE translation as a back-of-envelope mapping to Jones–Klenow (2016). Under $\lambda = 0.30$ (§7.5), the CE translation rises to $\sim 28\%$.

verdict is conditional on equal weights. A natural country-level extension lets $g^{US} \neq g^{EU}$ – different fairness beliefs across countries (Alesina–Angeletos 2005) – which we flag as an extension and do not pursue.

6.9 Verdict

Q7 verdict (canonical RTM + leisure-bonus culture). Europe is welfare-*dominant* on the labour-market block ($\Delta W = +0.082$, 27% above US joint surplus): the cultural premium ($\theta^{EU} = 0.156$) directly raises EU welfare by the leisure-bonus mechanism, and the canonical RTM union channel is roughly neutral on average under equal weights. **The substantive result is path-dependence, not the Shapley number.** Path A (culture, then union): culture +0.064, union +0.018; Path B (union, then culture): union −0.017, culture +0.100. The two paths agree culture is welfare-improving but disagree on the sign of the union-only effect, because under EU culture the firm-power baseline over-extracts hours and the union corrects it, while under US culture the union pushes h below the joint optimum. The Shapley average is a convention; the economic content is that **the union's welfare sign is conditional on the cultural primitive**. **Under realistic Saez–Stantcheva (2016) weights $\lambda = 0.30$ (§7.5)**: $\Delta W^{\lambda} = +0.154$, Shapley culture +0.079, union +0.076 – the union becomes a first-order contributor once redistribution is valued. The dominant US–EU welfare differences economy-wide come from non-labour channels (life expectancy, consumption, inequality; Jones–Klenow 2016); the labour-market block is sign-determinate but quantitatively second-order.

References

Acemoglu, D., & Pischke, J.-S. (1999). The Structure of Wages and Investment in General Training. *QJE* 114(1) [**Q5 inter-union competition**].

Aguiar, M., & Hurst, E. (2007). Measuring Trends in Leisure: The Allocation of Time Over Five Decades. *QJE* 122(3) [**Q1 time-use measurement**].

Alesina, A., & Angeletos, G.-M. (2005). Fairness and Redistribution. *AER* 95(4) [**Q7 cultural channel**].

Alesina, A., Glaeser, E., & Sacerdote, B. (2001). Why Doesn't the United States Have a European-Style Welfare State? *BPEA* 2001(2) [**Q7 cultural channel**].

Alesina, A., Glaeser, E., & Sacerdote, B. (2005). Work and Leisure in the United States and Europe. *NBER Macro Annual* 20 [**Q7 cultural channel**].

Alesina, A., & Giuliano, P. (2015). Culture and Institutions. *JEL* 53(4) [**Q7 cultural channel**].

Becker, G. S., Philipson, T. J., & Soares, R. R. (2005). The Quantity and Quality of Life and the Evolution of World Inequality. *AER* 95(1) [**Q7 life-expectancy welfare**].

Bénabou, R., & Tirole, J. (2006). Belief in a Just World and Redistributive Politics. *QJE* 121(2) [**Q7 cultural channel**].

Bewley, T. F. (1999). *Why Wages Don't Fall During a Recession.* Harvard University Press [**★ §5.6 downward wage rigidity**].

Bick, A., Brüggemann, B., & Fuchs-Schündeln, N. (2019). Hours Worked in Europe and the US. *Scand. J. Econ.* 121(4) [**★ Q1 data**].

Bick, A., Fuchs-Schündeln, N., & Lagakos, D. (2018). How Do Hours Worked Vary with Income? Cross-Country Evidence and Implications. *AER* 108(1) [**Q2 two-margin labour supply (BFL)**].

Binmore, K., Rubinstein, A., & Wolinsky, A. (1986). The Nash Bargaining Solution in Economic Modelling. *RAND J. of Economics* 17(2) [**Q5 bargaining solution**].

Bisin, A., & Verdier, T. (2001). The Economics of Cultural Transmission and the Dynamics of Preferences. *JET* 97(2) [**Q7 cultural transmission**].

Brown, J. N., & Ashenfelter, O. (1986). Testing the Efficiency of Employment Contracts. *JPE* 94(3) [**Q5 efficient-bargaining test**].

Blanchard, O., & Wolfers, J. (2000). The Role of Shocks and Institutions in the Rise of European Unemployment. *Economic Journal* 110(462).

Boppart, T., & Krusell, P. (2020). Labor Supply in the Past, Present, and Future. *JPE* 128(1).

Bowles, S. (1998). Endogenous Preferences. *JEL* 36(1).

Cahuc, P., & Carcillo, S. (2014). The Detaxation of Overtime Hours: Lessons from the French Experiment. *IZA WP*.

Calmfors, L., & Driffill, J. (1988). Bargaining Structure, Corporatism and Macroeconomic Performance. *Economic Policy* 3(6) [**Q5 hump-shape result**].

Card, D., Cardoso, A. R., Heining, J., & Kline, P. (2018). Firms and Labor Market Inequality: Evidence and Some Theory. *JOLE* 36(S1) [**Q4 modern monopsony evidence**].

Chemin, M., & Wasmer, E. (2009). Using Alsace–Moselle Local Laws to Build a DiD on the 35-Hour Workweek Regulation in France. *JLE* 27(4).

Chetty, R. (2012). Bounds on Elasticities with Optimization Frictions. *Econometrica* 80(3) [**Q2 frictions and Hicksian bounds**].

Diamond, P., & Saez, E. (2011). The Case for a Progressive Tax: From Basic Research to Policy Recommendations. *JEP* 25(4) [**Q2 top-rate formula**].

Chetty, R., Guren, A., Manoli, D., & Weber, A. (2011). Are Micro and Macro Labor Supply Elasticities Consistent? *AER P&P* 101(3) [**Q2 elasticity reconciliation**].

Cooley, T., & Prescott, E. (1995). Economic Growth and Business Cycles. In *Frontiers of Business Cycle Research*, Princeton.

Erosa, A., Fuster, L., & Kambourov, G. (2016). Towards a Micro-Founded Theory of Aggregate Labour Supply. *ReStud* 83(3) [**Q2 life-cycle aggregation**].

Estevão, M., & Sá, F. (2008). The 35-Hour Workweek in France: Straightjacket or Welfare Improvement? *JEEA* 6(5).

Hall, R. E., & Jones, C. I. (2007). The Value of Life and the Rise in Health Spending. *QJE* 122(1) [**Q7 VSL**].

Hansen, G. D. (1985). Indivisible Labor and the Business Cycle. *JME* 16(3).

Harsanyi, J. C. (1955). Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility. *JPE* 63(4) [**Q5 weighted-utilitarian foundation**].

Heathcote, J., Storesletten, K., & Violante, G. L. (2017). Optimal Tax Progressivity: An Analytical Framework. *QJE* 132(4) [**Q2 progressivity channel; §5.5b λ calibration**].

Holden, S. (1994). Wage Bargaining and Nominal Rigidities. *Scand. J. Econ.* 96(2) [**★ §5.6 bargained-wage rigidity**].

Jones, C. I., & Klenow, P. J. (2016). Beyond GDP? Welfare across Countries and Time. *AER* 106(9) [**Q7 benchmark**].

Layard, R., Nickell, S., & Jackman, R. (1991). *Unemployment: Macroeconomic Performance and the Labour Market.* Oxford University Press [**Q5 textbook**].

Lee, D., & Saez, E. (2012). Optimal Minimum Wage Policy in Competitive Labor Markets. *JPubE* 96(9-10) [**Q4 quantity instruments**].

MaCurdy, T., & Pencavel, J. (1986). Testing Between Competing Models of Wage and Employment Determination. *JPE* 94(3) [**Q5 right-to-manage**].

Manning, A. (1987). An Integration of Trade Union Models in a Sequential Bargaining Framework. *Economic Journal* 97(385).

Manning, A. (2003). *Monopsony in Motion: Imperfect Competition in Labor Markets.* Princeton University Press [**§4 firm-power baseline**].

Manning, A. (2021). Monopsony in Labor Markets: A Review. *ILR Review* 74(1) [**Q4 monopsony review**].

McDaniel, C. (2011). Effective Labor and Capital Income Tax Rates: 1950–2003. *Mimeo*, ASU [**★ Q1 panel (b)**].

McDonald, I. M., & Solow, R. M. (1981). Wage Bargaining and Employment. *AER* 71(5) [**Q5 efficient bargaining**].

Nash, J. F. (1950). The Bargaining Problem. *Econometrica* 18(2).

Oswald, A. J. (1985). The Economic Theory of Trade Unions: An Introductory Survey. *Scand. J. Econ.* 87(2) [**Q5 union-objective taxonomy**].

Piketty, T. (1995). Social Mobility and Redistributive Politics. *QJE* 110(3) [**Q7 cultural channel**].

Prescott, E. C. (2004). Why Do Americans Work So Much More than Europeans? *FRB Mpls Quarterly Review* 28(1).

Rogerson, R. (1988). Indivisible Labor, Lotteries and Equilibrium. *JME* 21(1).

Rogerson, R. (2008). Structural Transformation and the Deterioration of European Labor Market Outcomes. *JPE* 116(2) [**Q1 home-production extension**].

Rogerson, R., & Wallenius, J. (2009). Micro and Macro Elasticities in a Life Cycle Model with Taxes. *JET* 144(6) [**Q2 life-cycle aggregation**].

Saez, E. (2001). Using Elasticities to Derive Optimal Income Tax Rates. *ReStud* 68(1) [**Q2 elasticity-tax bridge**].

Saez, E., & Stantcheva, S. (2016). Generalized Social Marginal Welfare Weights for Optimal Tax Theory. *AER* 106(1) [**Q7 fairness-weighted welfare**].

Stevenson, B., & Wolfers, J. (2008). Economic Growth and Subjective Well-Being: Reassessing the Easterlin Paradox. *BPEA* Spring 2008 [**Q7 SWB residual**].

Replication scripts in `research/`*:* `build_figure_panel.py` *(Q1 figure);* `sim_models_v2.py` *(Q2 decomposition, Q5 RTM, §5.5b Saez–Stantcheva weights, §5.6 sectoral shocks, Q7 cultural expansion + layered framework, §7.5 sensitivity);* `sim_mechanisms.py` *(§4 cap simulation).*